Gemini Event Consolidation Evaluation

StockEvent Lens | NVIDIA financial news | Final completed run, 25 batches / 125 articles

125 / 125	94.0%	88.9%	88.0%
Articles processed	Grouping F1	Label macro-F1	Label accuracy

Classification performance



Class	Precision	Recall	F1	TP / FP / FN
Repeated event	95.7%	88.0%	91.7%	22 / 1 / 3
Updated event	79.2%	76.0%	77.5%	19 / 5 / 6
Conflicting event	100.0%	88.0%	93.6%	22 / 0 / 3
Different event	86.2%	100.0%	92.6%	25 / 4 / 0

Key findings

1. Strong event grouping. Pairwise precision was 100.0%, recall 88.7%, and F1 94.0%. The pipeline rarely merged unrelated article pairs, but missed some true relationships.

2. Updates are the main weakness. Update F1 was 77.5%, below repeat (91.7%), conflict (93.6%), and different-event (92.6%). Earlier logs show some update articles received no candidate from retrieval, so this is not solely an LLM judgment problem.

3. Reliability recovery worked. Four complete batches were preserved from the original run; 21 batches were recovered. The final report has zero processing errors, although two classifications were marked uncertain.

Evaluation methodology and interpretation

Each of 25 independent five-article batches contains an original/reference article and labeled repeat, update, conflict, and different-event cases (with some article order variation). Embedding-based candidate retrieval precedes Gemini judgment. The original article is not scored as a four-class relationship; the reported label accuracy covers 100 scored cases. The reported 88.0% accuracy is 88/100; two cases are uncertain. This is an evaluation on the team's labeled dataset, not a claim of real-world reliability.

Operational summary

Model: gemini-3.8-flash | 77 model calls | Tokens: 179,767 input / 50,380 output | 25/25 batches, no processing errors.

Source: final terminal log. Detailed results: evaluation/consolidation/results/gemini_125_completed.json